

Prime Dynamics Program, Volume III

Deterministic Numerator Anchors, Analytic Tails, and Counterloops

A Provenance-Preserving Synthesis of RH-242–RH-281

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Abstract

This volume synthesizes RH-242–RH-281, the forty-paper interval in which the prime-dynamics program separates an exact deterministic target from the still-unidentified moving noisy cloud. The deterministic side advances from graded superloop identities and coefficient dictionaries to a certified all-order envelope

$$|a_n| < 48q_*^n, \quad q_* = 0.7008752258547757 \dots, q_{quad}n \geq 2,$$

the sharp limit $a_n/q_*^n \rightarrow 1$, the exact radius $\rho_* = 1.4267874838640739 \dots$, and a resolution-clocked monodromy counterloop with a sharp minimal-rank theorem.

The volume records two complementary negative lessons. First, several frozen, idempotent, fractional, and unit-cap selector classes fail in their declared finite algebras. Second, a complete root-of-unity shell can erase any fixed finite moment prefix while retaining a freely scalable next moment. Thus finite coefficient matching does not imply a moving all-order cloud envelope, even when the deterministic target envelope is sharp.

We prove an executable provenance statement, a deterministic closure theorem relative to the upstream sources, and a finite-head separation theorem. The actual cloud identification remains open: no source supplies a legal operator-derived head, a cloud-to-anchor coefficient bridge, aggregate noisy-cloud transport, or an instantiated variable-rank quotient theorem. Gates A–E remain false/open. This synthesis does not construct a Hilbert–Polya operator, identify Riemann zeros, prove a von Mangoldt trace, prove completed-zeta divisor equality, or prove the Riemann Hypothesis.

Keywords: deterministic numerator; spectral anchors; regularized determinants; analytic tails; counterloops; coefficient envelopes; provenance audit.

1 Scope, data types, and status words

Volume III occupies the interval between the physical-cloud frontier of Volume II and the actual noisy-head analysis of Volume IV. The position is important: a deterministic target can be solved exactly without being identified with an actual noisy operator.

The numbered papers remain atomic sources. We label a statement **PROVED** only in its source scope, **CERTIFIED** when it is a finite exact or outward computation, **CONDITIONAL** when its implication has open physical hypotheses, and **SCOPED NEGATIVE** when it excludes one named class rather than every possible selector or operator.

The four phase endpoints are RH-251, RH-261, RH-271, and RH-281. Each reviews the preceding ten-paper block and fixes its route coordinate. No finite ledger count is promoted to an all-order theorem unless an analytic source paper supplies that theorem.

2 Executable source inventory

Theorem 2.1 (Volume III provenance theorem). *The executable inventory finds exactly one complete source directory for each integer $242 \leq n \leq 281$, assigns the forty labels to four consecutive ten-paper phases, and binds selected source files by SHA-256. The four review anchors are RH-251, RH-261, RH-271, and RH-281.*

Proof. The builder discovers directories by their numerical prefix, rejects a duplicate or incomplete source, compares the resulting labels with the full integer interval, and checks that the declared phase union is the same interval. It hashes available README, TeX, theorem-ledger, roadmap, and result records. The independent verifier replays these hashes. This proves source presence and identity only. $\square$

3 RH-242–RH-251: superloops, quotients, and frozen anchors

RH-242 gives a graded periodic-superloop representation. RH-243 supplies a deterministic numerator coefficient dictionary. The early block then asks whether a finite cloud can be selected so that its moments realize the required anchored prefix while the quotient tail remains controlled.

Several exact structural tools survive the audit. Orthogonal quotient grouping gives a trace identity; a block-power theorem gives a sufficient uniform-envelope criterion; and finite-head/analytic-tail gluing isolates the necessary obligations. On the negative side, separately majorizing physical and atomic sectors incurs a superunit rate. The complete single-use frozen shell zonotope has zero passing endpoints in the archived class, with dual certificates. Unbounded nonnegative weights create formal fits only by using pathological magnitudes and do not define legal spectral multiplicities.

RH-251 therefore records an honest mixed verdict: exact superloop and quotient algebra, but no legal anchored head and no complete current certificate. This is a scoped obstruction to the frozen class, not a proof that every invariant selector fails.

4 RH-252–RH-261: analytic tails and selector legality

This phase adds a genuinely analytic target-side input. The deterministic numerator has an exact Cauchy tail interface on every disk

$$0 \leq R < S < r_H \lambda,$$

where $r_H \lambda = 1.42678748386407\dots > 1$. At this point the boundary supremum was not yet certified numerically; later papers close it.

The deterministic anchor atlas reaches order 28, but its fitted root rate is descriptive finite data. Enlarging every resolved spectral window by sixteen roots still produces no legal anchored head in the archived single-use box. Real conjugate-closed polynomial idempotents collapse to binary generalized-root-space masks. Arbitrary fractional signed fits have nontrivial monodromy unless their combined exponents are integral, and the first monodromy-legal lattice $\{-1, 0, 1\}$ again fails in the archived class.

The quotient diagnostic finds contractive twelfth powers at the tested endpoints but lacks nine endpoints and a continuum bridge. Its finite worst root rate worsens rather than stabilizes. The RH-261 review accordingly separates one proved analytic obligation from four unproved physical or uniform obligations.

5 RH-262–RH-271: deterministic all-order closure

The next block closes the target side. It certifies the boundary constant, proves an exact all-order parity anchor, controls the direct order-29 tail, and constructs a finite ladder leading to the global estimate

$$|a_n| < 48q_*^n, qquad n \geq 2, qquad q_* = 0.7008752258547757 \dots \quad (1)$$

The coefficient base is sharp:

$$\frac{a_n}{q_*^n} \longrightarrow 1, \quad \rho_* = q_*^{-1} = 1.4267874838640739 \dots \quad (2)$$

Theorem 5.1 (Deterministic closure theorem). *In the deterministic numerator normalization fixed by RH-262–RH-268, the all-order parity anchor, envelope (1), sharp asymptotic law (2), and exact radius ρ_* are proved. These conclusions require neither a finite-order fit nor an extrapolated noisy spectrum.*

Proof. The statement is the conjunction of the exact source theorems bound by the RH-271 review: the certified boundary constant supplies the analytic input, the parity anchor fixes all coefficient orders, the factorwise tail closes the finite transition, and the sharp base theorem supplies the limit and radius. The inventory binds those sources. No actual-cloud identification is included in the hypotheses or conclusion. $\square$

The phase also proves why this closure is not enough. A complete shell of $(N + 1)$ -st roots of unity has trace moments zero through order N while its moment at order $N + 1$ may be scaled. This exact separation survives any fixed finite prefix.

Theorem 5.2 (Finite-head separation theorem). *For every finite order bound N , equality of two spectral moment lists through order N does not determine a uniform all-order envelope or the next moment without an additional structural hypothesis.*

Proof. Adjoin a complete root-of-unity shell of size $N + 1$. Orthogonality makes its power sums vanish for orders $1, \dots, N$, while the $(N + 1)$ -st power sum is nonzero and scales with the shell amplitude or multiplicity allowed by the information class. The finite prefix is unchanged and the next coefficient is free. This is a logical counterexample; it does not claim that the archived noisy quotient contains the shell. $\square$

6 RH-272–RH-281: counterloops and quotient limitations

RH-272 constructs a resolution-clocked monodromy counterloop giving an exact all-order coefficientwise deterministic bridge. RH-273 proves its minimal rank and equality-case uniqueness. RH-274 identifies aggregate Fourier control as the correct noisy-cloud input and gives a maximum-phase counterexample to weaker coordinatewise reasoning. The seven-row RH-275 audit does not certify that aggregate input.

The operator branch supplies complementary negative and positive facts. RH-276 proves that raw Hilbert–Schmidt mass diverges like $\sigma^{-1/2}$. RH-277 excludes fixed-rank Calkin contraction in the natural stationary L^2 geometry at zero noise. RH-278 shows that positive-noise local shell charts can activate the earlier quotient theorem. RH-279 proves a variable-rank block-power tail criterion, but its physical hypotheses are uninstantiated. RH-280 and RH-281 keep the deterministic and spectral ledgers separate.

Proposition 6.1 (Counterloop type boundary). *The resolution-clocked counterloop is an exact deterministic all-order object. Its moments cannot be substituted for an actual noisy head unless a separate theorem proves aggregate noisy-cloud transport and canonical identification on the same clock.*

Proof. The source construction of the counterloop uses a graded monodromy model, whereas the open ledger asks for an actual noisy spectral aggregate. Equality of notation or coefficient length does not supply a map between these data types. RH-274 identifies the missing aggregate Fourier hypothesis and RH-275 does not certify it. $\square$

7 Selector and quotient firewalls

The volume supports several precise stopping rules.

1. Do not rerun the same frozen shell window with unbounded weights and call the result a legal spectral head.
2. Do not interpret complex polynomial coefficients as fractional spectral multiplicities; idempotence and monodromy impose exact algebraic constraints.
3. Do not promote a finite list of contractive block powers to a uniform small-noise theorem.
4. Do not infer a moving coefficient bridge from any fixed finite anchor, even when the deterministic target envelope is sharp.
5. Do not use raw Hilbert–Schmidt compactness at zero noise; its mass has the proved divergent scale.
6. Do not extend the fixed-rank Calkin no-go to every variable-rank or adapted geometry.

These rules preserve both sides of the result. The deterministic theorem is stronger than a fit, while each negative conclusion is narrower than global nonexistence.

8 The surviving physical obligations

A complete physical bridge requires at least:

- (i) a legal operator-derived anchored head;
- (ii) an exact cloud-to-deterministic coefficient bridge;
- (iii) aggregate noisy-cloud transport into the monodromy shell;
- (iv) a common-space ideal estimate and moving isolating contour; and
- (v) an instantiated contractive power for the variable-rank quotient.

The all-order target tail and deterministic counterloop do not replace any of (i)–(v). Conversely, the scoped selector failures do not prove that a new invariant quotient or physical construction cannot satisfy them. The current coordinate is

deterministic_all_order_closed_actual_cloud_identification_open.

9 Reproducibility and claim boundary

The volume audit discovers all forty sources, checks the four phase ranges, hashes selected upstream files, stores the exact deterministic constants as strings, and records every Gate and forbidden macro claim as false. Tests independently enforce the source range, the deterministic/physical split, and the route coordinate. Archive generation then hashes the manuscript, tests, scripts, result records, and PDF before an independent replay.

The audit checks identity and declared status; it does not re-prove forty source papers. Finite selector counts remain finite certificates. The actual cloud identification remains open. Gates A–E remain false/open. This volume does not construct a Hilbert–Polya operator, identify Riemann zeros, prove a von Mangoldt trace, prove equality with the completed-zeta divisor, or prove the Riemann Hypothesis.

A Atomic source index

This machine-generated table lists every canonical numbered source in Volume III. It is a provenance locator, not a uniform theorem-status assignment.

source	canonical directory
RH-242	RH-242-cloud-extracted-periodic-superloop-representation
RH-243	RH-243-deterministic-numerator-coefficient-anchor-dictionary
RH-244	RH-244-anchored-shell-prefix-availability-obstruction
RH-245	RH-245-orthogonal-quotient-superloop-compression
RH-246	RH-246-block-power-quotient-envelope-criterion
RH-247	RH-247-separate-absolute-majorant-barrier
RH-248	RH-248-anchored-shell-zonotope-reachability-obstruction
RH-249	RH-249-unbounded-shell-multiplicity-pathology
RH-250	RH-250-anchored-head-analytic-tail-gluing-criterion
RH-251	RH-251-ten-layer-superloop-anchor-frontier-review
RH-252	RH-252-deterministic-numerator-analytic-tail-certificate
RH-253	RH-253-extended-deterministic-anchor-atlas
RH-254	RH-254-expanded-resolved-candidate-window-atlas
RH-255	RH-255-expanded-window-anchored-zonotope-obstruction
RH-256	RH-256-invariant-polynomial-selector-binary-collapse
RH-257	RH-257-monodromy-integrality-barrier-for-signed-moment-fits
RH-258	RH-258-unit-cap-signed-integer-selector-obstruction
RH-259	RH-259-extended-quotient-block-power-diagnostic
RH-260	RH-260-updated-anchored-head-tail-certificate-ledger
RH-261	RH-261-ten-layer-analytic-tail-selector-frontier-review
RH-262	RH-262-certified-deterministic-numerator-boundary-budget
RH-263	RH-263-parity-resolved-deterministic-numerator-tail
RH-264	RH-264-direct-factorwise-deterministic-tail-certificate
RH-265	RH-265-certified-deterministic-tail-ladder
RH-266	RH-266-finite-sample-quotient-uniformity-obstruction
RH-267	RH-267-certified-unified-deterministic-trace-envelope
RH-268	RH-268-sharp-deterministic-coefficient-radius-law
RH-269	RH-269-contour-stable-uniform-quotient-criterion
RH-270	RH-270-updated-deterministic-envelope-quotient-certificate-ledger
RH-271	RH-271-ten-layer-deterministic-envelope-quotient-frontier-review
RH-272	RH-272-resolution-clocked-monodromy-counterloop-bridge
RH-273	RH-273-minimal-rank-pole-counterloop-uniqueness
RH-274	RH-274-fourier-defect-cloud-bridge-criterion
RH-275	RH-275-archived-cloud-fourier-defect-audit
RH-276	RH-276-small-noise-hilbert-schmidt-mass-obstruction
RH-277	RH-277-fixed-rank-calkin-quotient-contraction-obstruction

source	canonical directory
RH-278	RH-278-positive-noise-s2-analytic-shell-transport
RH-279	RH-279-variable-rank-block-power-tail-criterion
RH-280	RH-280-dual-counterloop-spectral-ledger
RH-281	RH-281-ten-layer-counterloop-quotient-frontier-review